

Calc Formula Engine Calculation Tes

Status	Done!
Duration	0.00

WARNING!

Depending on the speed of your hardware, the recalculation process may from 30 seconds to over a minute, during which time Calc will become too slow to use. In order to be able to operate the test, please do not forget to switch Macro Recording on. The document must not be read-only, use “Edit File” toolbar button in case of read-only document.

Problem Description

This document contains three data sheets containing sets of daily groundwater different sites: one located in Charleston, South Carolina, one located in Hilton one located in Orlando, Florida. All data used in this document are publicly available from the groundwater data archive website of the US Geological Survey.

Designation	Date	Level	Elevation	Average	Delta Low
D27	1990-04-03	5.29	2.21	-55.22	-98.3
D27	1990-04-04	5.33	2.17	-55.18	-98.3
D27	1990-04-05	5.52	1.98	-54.99	-98.3
D27	1990-04-06	5.70	1.80	-54.81	-98.3
D27	1990-04-07	5.85	1.65	-54.66	-98.3
D27	1990-04-08	5.89	1.61	-54.62	-98.3
D27	1990-04-09	6.00	1.50	-54.51	-98.3
D27	1990-04-10	6.08	1.42	-54.43	-98.3
D27	1990-04-11	6.17	1.33	-54.34	-98.3

D27	1990-04-12	6.23	1.27	-54.28	-97.9
D27	1990-04-13	6.29	1.21	-54.22	-97.9
D27	1990-04-14	6.08	1.42	-54.43	-98.9
D27	1990-04-15	6.08	1.42	-54.43	-98.9
D27	1990-04-16	6.14	1.36	-54.37	-98.9



On each data sheet, column A:K contain raw data, of which the only data used is column K. Column K contains daily depths to the water table below ground surface as measured. These depths are converted into elevations by subtracting them from the known ground surface elevation in cell L2. Column M shows the difference from the average elevation over the entire period. Column N and O show the differences from the lowest and highest elevations over the entire period, respectively.

When pressing the "Re-calculate" button, it forces all the formula cells present in the spreadsheet to be recalculated, that is, columns L through O in the two data sheets will go through a full recalculation. Once the recalculation is finished, the Status cell will show Finished and the Duration cell will show the number of seconds for the whole re-calculation has taken to complete.

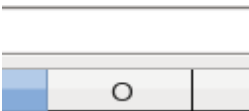
Proposed Solution

We will make two changes to Calc core in order to speed up the recalculation. First, we will change Calc's cell storage to store their raw values in arrays, and modify the formula in a cell so that the entire array can be fetched in one step rather than on a per-cell basis. Secondly, we will modify the formula parser to identify the range of identical formula tokens and use that information during recalculation to recalculate those cells in that range in one step using GPU. We will only track identical formulas within the same column.

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levels measured at three
Head, South Carolina, and
ilable, and are pulled directly



	High
92	3.48
88	3.52
69	3.71
51	3.89
36	4.04
32	4.08
21	4.19
13	4.27
04	4.36

98	4.42
92	4.48
13	4.27
13	4.27
07	4.33

in this calculation is column measured. Column L converts and surface elevation located entire measurement period. over the entire measurement

in this document to be fully re- calculation. When the re- shows the total time in

speed. Firstly, we will refactor interpreter so that the values in will modify the formula storage recalculation time to calculate nula token spans that are in

Input	Sub	7.5 Add	8.5 Mul	3.5	COUNT+SUM-Avg*COUNT
5.29		2.21	13.79	18.515	14.0005239
5.33		2.17	13.83	18.655	13.8954543
5.52		1.98	14.02	19.32	13.41716874
5.7		1.8	14.2	19.95	12.99346867
5.85		1.65	14.35	20.475	12.66030281
5.89		1.61	14.39	20.615	12.57432452
6		1.5	14.5	21	12.34379524
6.08		1.42	14.58	21.28	12.18137688
6.17		1.33	14.67	21.595	12.00369067
6.23		1.27	14.73	21.805	11.8880853
6.29		1.21	14.79	22.015	11.77468544